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Computational and experimental investigation of near-field sonic boom of a HTV-2 type hypersonic boost gliding vehicle

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Understanding and prediction of sonic boom are critical for the development of the next generation hypersonic vehicles. Low-boom design not only can minimize the thunderclap-like acoustic noise, but can enable higher propulsion efficiency as well. Before this project, hypersonic sonic boom measurements for slender bodies are not available in the literature. In this study, a Hypersonic Technology Vehicle 2 (HTV-2) type boost gliding vehicle (BGV) model, with high lift-to-drag ratio $L/D = 2.4 - 3$ (varying with the pitch angle), was designed and fabricated. An off-body pressure measurement device was fabricated for the desired sonic boom (i.e., over-pressure) signature measurement. The first set of hypersonic sonic boom shock tunnel tests was then conducted using the newly fabricated platform. The tests were run with different Mach numbers and pitch angles. Computational fluid dynamic (CFD) simulations were well validated by the shock tunnel test data. Using both fully laminar and fully turbulent (using the Spalart-Allmaras (SA) Reynolds-averaged Navier–Stokes (RANS) model) CFD simulations, potential laminar-to-turbulent transition locations were identified based on the agreement between the laminar or turbulent simulation and the surface sensor measurement of both heat transfer and pressure. CFD simulation indicated that the potential transition has very limited impact on the peak over-pressure, but has significant impact on the tail of the signature. CFD prediction falls within the uncertainty ranges of the off-body pressure measurement data, but the measurement uncertainty needs to be improved in future shock tunnel tests.

I. Nomenclature

r	=	Radius of sonic boom source cylinder
θ	=	Azimuthal angle, the angle between r axis and y axis
x	=	Cartisian coordinate along free-stream direction
y	=	Cartisian coordinate along gravity direction
z	=	Cartisian coordinate in horizontal direction
α	=	Flight pitch angle
p_∞	=	Free-stream pressure
Δp	=	Over-pressure with respect to p_∞: $\Delta p = p - p_\infty$
ρ_∞	=	Free-stream pressure density
U_∞	=	Free-stream velocity
M_∞	=	Free-stream Mach number
Re_∞	=	Free-stream unit Reynolds number: $Re_\infty = \rho_\infty U_\infty / \mu_\infty \times$ (unit length)
T_∞	=	Free-stream flow temperature
T_w	=	Wall temperature used as the boundary condition of CFD

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Re_θ	=	Reynolds number based on momentum thickness $Re_\theta = \rho U_\infty \delta_\theta/\mu$, where δ_θ is the momentum thickness
M_e	=	Edge Mach number evaluate by the velocity and speed of sound at the edge of the boundary layer
L	=	Body length of the test article

II. Introduction

Sonic boom is the audible shock wave produced by supersonic or hypersonic movers. Typically, sonic boom carries tremendous amount of energy and usually sounds like explosion or a thunderclap to a human ear. Unlike other airborne sounds (e.g., jet noise), sonic boom (i.e., normalized over-pressure $\Delta p/p_\infty = (p - p_\infty)/p_\infty$, where p_∞ is the free-stream pressure) exhibits thousands of times larger amplitude up to $\Delta p/p_\infty \sim O(1 - 100)$ and extreme frequencies up to $O(1 - 10)$ kHz near the mover. High-energy and high-frequency pressure wave associated with sonic boom allows it to overcome atmospheric dispersion and dissipation during the long-range travel and finally reach the ground with considerable loudness, even during the aircraft's cruising phase. This phenomenon has led to commercial supersonic flights being recognized as significant sources of overland noise pollution. Consequently, the Federal Aviation Administration (FAA) imposed a ban on overland commercial supersonic flight. Low-boom design is hence a practical need for any commercial supersonic aircraft. Furthermore, due to the substantial energy carried by the shock wave, low-boom design is also advantageous in an energy-saving perspective. This goal has seen success in controlling the shock strength at relatively low Mach numbers (i.e., free-stream Mach number $M_\infty < 2$) [1]. For example, as the revival of supersonic commercialization projects, NASA's QueSST project [2] came out and is ready for flight tests. In contrast, hypersonic sonic boom for lifting movers is lack of study, and the possibility of extending low-boom design to hypersonic aircrafts has not been explored.

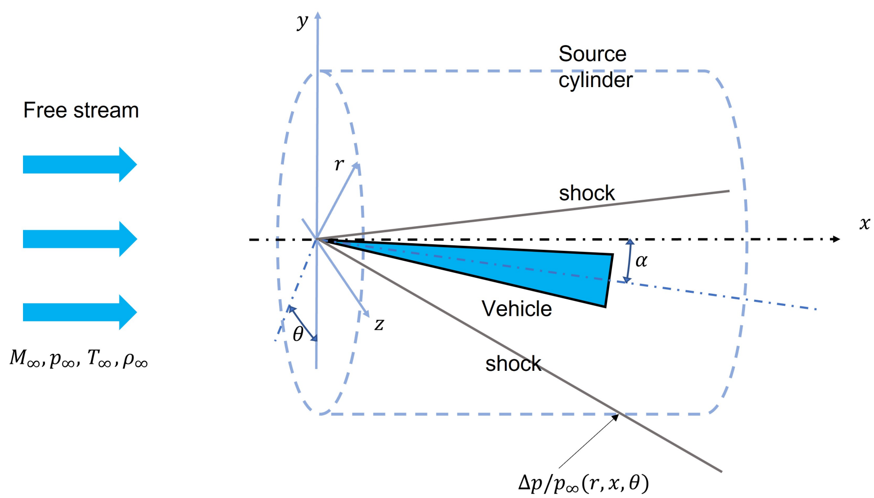

Fig. 1 Typical coordinate system for sonic boom problem. x is the Cartesian coordinate along free-stream direction, y is the Cartesian coordinate along gravity direction, r is the radial direction of the sonic boom source cylinder, and θ is the azimuthal angle between r axis and y axis, and α is the flight pitch angle. M_∞, p_∞, T_∞, and ρ_∞ are the free-stream Mach number, pressure, temperature, and density, respectively. $\Delta p = p - p_\infty$ is the over-pressure (i.e., sonic boom).

Currently, the sonic boom prediction is usually composed of two major components: near-field data sampling and propagation of data to middle and far fields. The near-field data, which is usually sampled at 1-4 body lengths in radial direction of the source cylinder (as shown in Fig. 1, marked by r), could be obtained from either wind/shock tunnel tests or CFD simulations. In the experimental near-field data sampling, the major technical challenge comes from the off-body measurement device. It is not easy to design a corruption-free thin wedge inside a supersonic/hypersonic flow domain. An improper design could result in shock reflection and corresponding signature distortion [3]. According to Morgenstern [4], the most recent successful design is a so-called "pressure rail" structure. Such a device, capable of minimizing the perturbation induced by the off-body measurement system, is used in this study. Regarding the CFD data sampling, the major challenge is the high resolution requirement near the shock front, and hence a tedious iteration of re-meshing is often needed. For the propagation of sonic boom signature to middle and far fields, if the input shock from the near-field data sampling is sufficiently weak, the well-established 1-D augmented burgers equation model [5], has been proven to have strong capability and excellent agreement with full-scale CFD simulations [6]. While the near-field shocks of supersonic movers are sufficiently weak for the propagation, the near-field shocks of hypersonic movers are too strong for the propagation and hence much larger near-field tunnel/computational domain sizes are required to relax the shocks to sufficiently weak before the propagation. There are several existing tools implemented the solver for augmented burgers equations together with the 1976 standard atmosphere profile [7], including PCBOOM3 [8] from Wyle Laboratories and US Air Force, and sBOOM [9] from NASA Langley Research Center. The recent practice of such systematic investigation of supersonic sonic boom could be found in three sonic boom prediction workshops hosted by NASA Langley [10–12].

According to Tiegerman [13], there is no difference in physical process between hypersonic sonic boom and supersonic sonic boom, when the pressure wave is weak enough, or equivalently, when r is large enough. For this reason, the key to hypersonic sonic boom prediction is to sample the near-field data by either CFD or experiment. Limited by the capability of facilities, the hypersonic sonic boom data are mainly ground records for sphere-like hypersonic movers such as reentry capsules [14] and meteors [15]. The study of hypersonic sonic boom for slender bodies are still blank, and the sonic boom database for lift-dominant (i.e., lift-to-drag ratio $L/D > 1$) hypersonic movers are not established. For this reason, we launched a computational and experimental investigation of near-field hypersonic sonic boom for a Hypersonic Technology Vehicle 2 (HTV-2) type boost gliding vehicle (BGV) geometry [16] with L/D varies from about 3 to about -2.4 for the pitch angle from $-5°$ to $5°$. Unlike the conventional hypersonic measurements which focuses on the near-body/boundary layer status (still important for CFD validation), sonic boom measurements focus on the off-body pressure signatures. Our off-body measurement device was fabricated following the design of "pressure rail" [17]. The data is serving as a key outcome of our hypersonic sonic boom project and the validation data of our computational and theoretical model development [18].

The remaining part of the paper is organized as: In Sec. III, we will brief introduce the fabrication of the test article and measurement platform, and the CFD simulation tools. In Sec. IV, we will validate the CFD prediction by measurement data from various perspective, including force balance measurement, on-body heat transfer and pressure measurement, shock angle, and off-body pressure signature. The uncertainty of the measurement will also be identified. In Sec. V, a brief summary was made for the key findings of this study and required improvements in the future plan.

III. Methodology

A. Test platform build up

1. Model fabrication

Based on the HVT-2 type BGV geometry simulated by Niu et al. [16], four variations with differences in shape and nose radius were generated, and the sketches of these 4 candidates with the sting holders are given in Fig. 2. . To maximize the potential lift effects on sonic boom, CFD simulations are performed to evaluate the possible lift-to-drag ratio (L/D), and nose radius is used to control the drag (D). The nominal tunnel condition $p_\infty = 973$ Pa, $T_\infty = 300$ K, and $\rho_\infty = 0.0113$ kg/m^3 are used in this series of test runs with pitch angle $\alpha = 0$ and free-stream Mach number $M_\infty = 5$ is used. The computed forces and L/D are given in Table 1. Based on the maximum L/D obtained from CFD, model 0004 and 0006 are better than 0003 and 0005. Considering the BGV size and the available space in CUBRC's LENS 48-inch Shock Tunnel, 0004 was finally chosen, because of its smaller space occupancy compared with 0006.

Table 1 Lift, drag, and lift-to-drag ratio (L/D) of the four candidate models: data sampled at $M_\infty = 5$ with a nominal tunnel condition of $p_\infty = 973$ Pa, $T_\infty = 300$ K, and $\rho_\infty = 0.0113$ kg/m^3.

Model	Lift (N)	Drag (N)	L/D
0003	40.25	18.15	2.22
0004	40.00	16.08	2.49
0005	39.66	18.59	2.13
0006	83.22	33.91	2.45

The sensitivity of lift-to-drag ratio L/D to pitch angle α and free-stream Mach number M_∞ is then tested to model

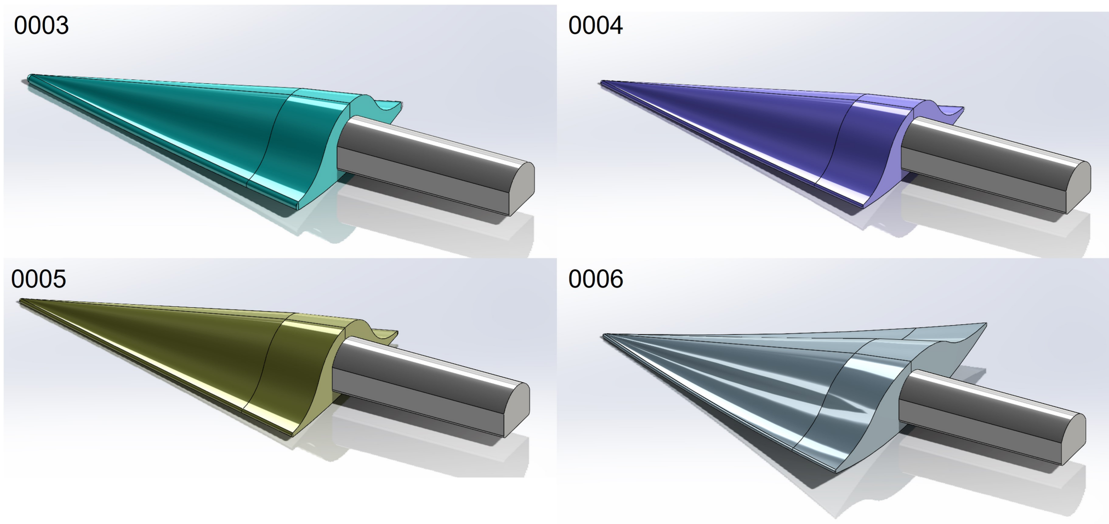

Fig. 2 Computer-aided design (CAD) view of all of the four candidate BGV geometries.

0004. As indicated by Fig. 3, L/D is not sensitive to the free-stream Mach number M_∞, but it varies significantly from 3 to -2.4 when pitch angle α changed from $\alpha = -5^\circ$ to 5°.

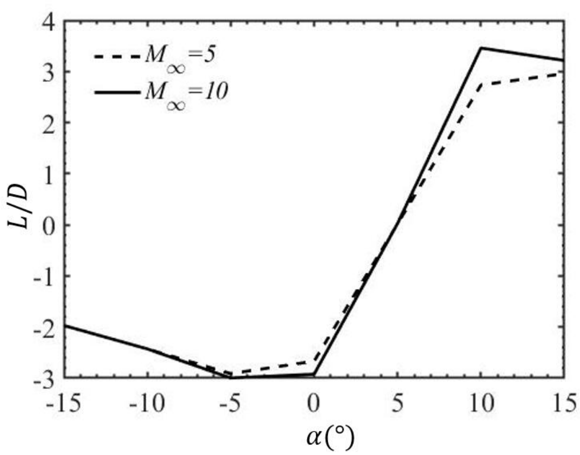

Fig. 3 Lift-to-drag ratio L/D of model 0004 as a function of pitch angle α at two different M_∞, with $p_\infty = 973$ Pa, $T_\infty = 300$ K, and $\rho_\infty = 0.0113$ kg/m^3 held as constants. The sign of the ratio indicates the direction of the lift.

The completed design is shown in the Fig. 4 by scaling up model 0004 by a factor of 4. The final size is 14.7 inch in length and 9.02 inch in wide. One of CUBRC's high aspect ratio balances has been selected for the test, because it is most compatible with the geometry. The test article has limited surface instrumentation as shown in Fig. 5, with red dots indicating heat transfer (T) sensors and green dots indicating pressure (P) sensors on the test article surface. Black dots indicate the fiducial points on the surface for the balance calibration (C) process. For a light-weight and rigid structure, magnesium are used for the test article. The balance is constructed to be a very stiff assembly, which has been previously used for short-duration force measurements at CUBRC. It is a 6-component balance designed around sensors, ideal for the short duration tests performed in CUBRC's LENS 48" Shock Tunnel.

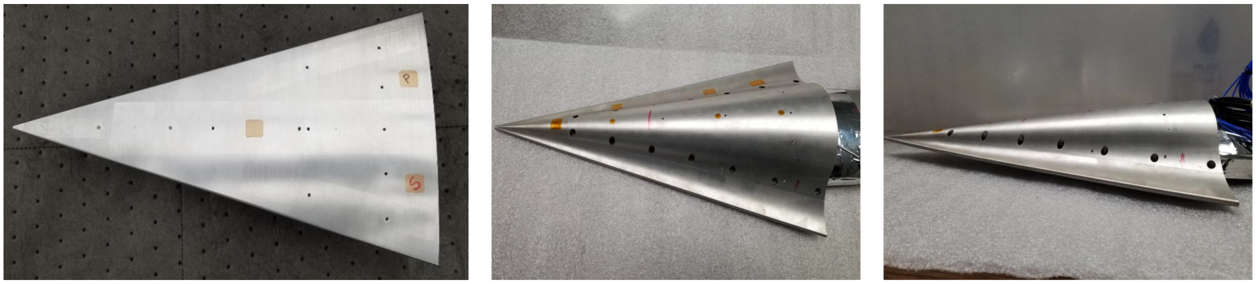

Fig. 4 Fabricated test article with sensors installed. Left: bottom view; Middle: top view; Right: side view.

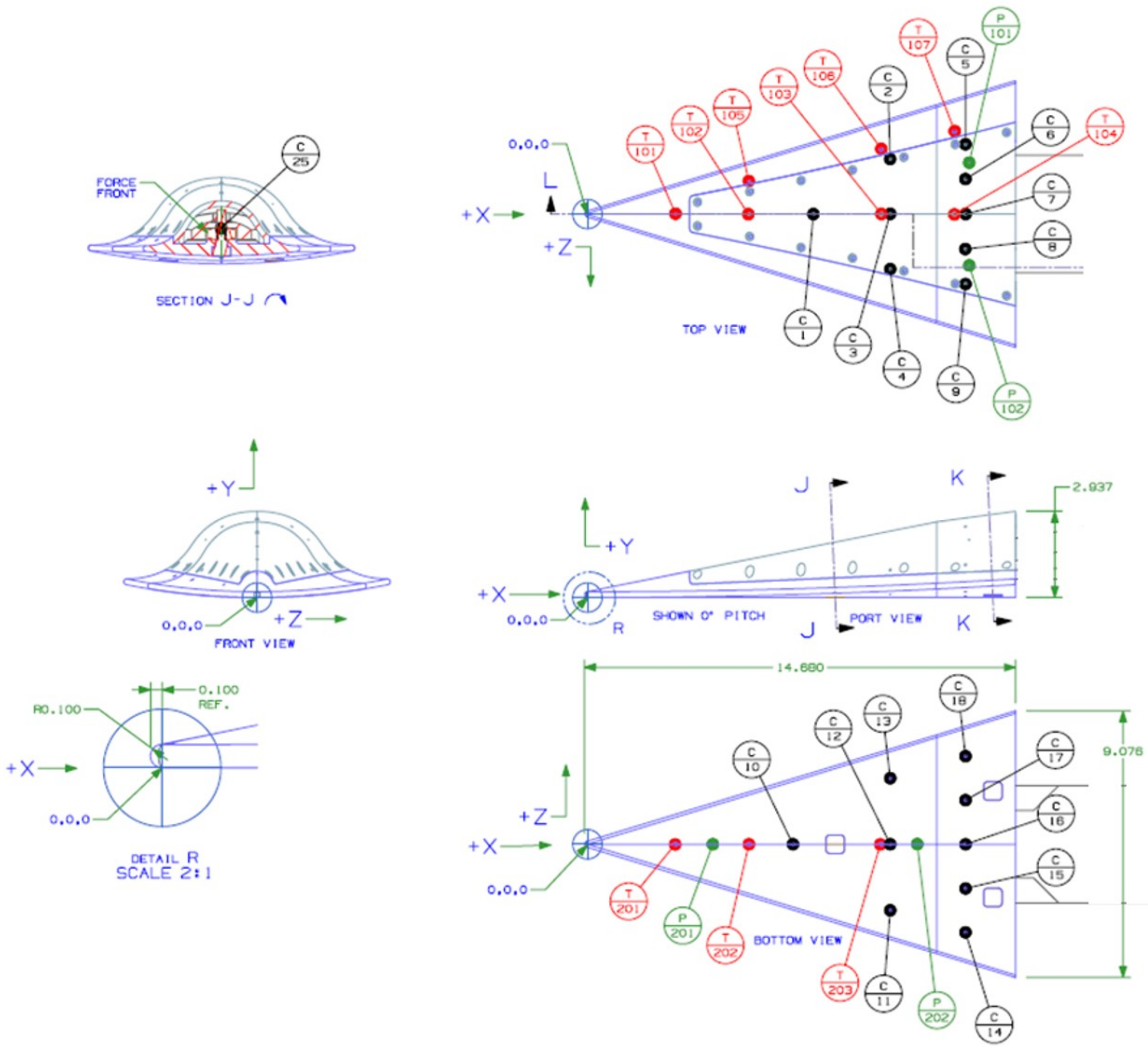

Fig. 5 CAD design of the BGV test article and sensor locations: red dots are the heat transfer (T) sensors, green dots are the pressure (P) sensors, and black dots are the fiducial points on the surface for the balance calibration (C) process.

After the finalization of the test article, the model for shock tunnel tests have been fabricated. Its surface pressure sensors and heat transfer sensors have been installed. For a lightweight, rigid structure, magnesium is the material of choice for the fuselage and cover. The photos of the BGV test article are shown in Fig. 4.

2. Measurement platform compilation

For sonic boom, high quality off-body pressure measurement is very critical. As indicated by Morgenstern [4], the key of the success design of a sonic boom pressure rail is to minimize the perturbation introduced by the rail plate. This logic results in a thin plate with a width of ~ 1.3 inch, as shown in Fig. 6(a). Taking the advantage of flexibility of the mounts in the spanwise direction, the plate is allowed to move laterally for up to 4", such that off-centerline pressure signal can be also captured to evaluate the azimuthal angle θ distribution of the sonic boom signature.

As indicated in Fig. 6(a), there are 20 pressure gauge drilled and installed. Although the resolution is lower than typical test in supersonic sonic boom measurement practice, the shock structure generated by the test article in this study is a single N wave, which has much less fine structures to resolve due to the much more compact shock structure at hypersonic speeds, as shown in Fig. 6(b). For this reason, coarser resolution could still be capable to characterize the shock structure in the tests, because the absence of those confined spikes at low supersonic speeds (e.g., see Fig. 13 and Fig. 14 in Durston et al. [19]). The distribution of these pressure gauge slots on the off-body measurement plate (which is about 1 meter long) is given as: 70% of the sensors on the first half of the plate, and 30% on the second half; uniformly distributing the sensors on each half. This off-body pressure measurement panel was fabricated in the CUBRC machine shop, as shown in Fig. 6(c). Depending on the angle of the shock wave for each case, the plate may need to be put either more upstream (i.e., front) or more downstream (i.e., aft) in the axial direction. To enable this flexibility, four mounts were built in the shock tube tunnel to allow an up to 17.5" axial offset, as shown in the Fig. 6(d). The other concerns of sonic boom measurement including the reflection from the tunnel wall and the interference between the bulk flow and tunnel boundary layer. In this configuration, due to relative large standing-off distance of the plate from the tunnel wall, these two problems are avoided at the two free-stream Mach numbers (6.5 and 8.0) in the tests.

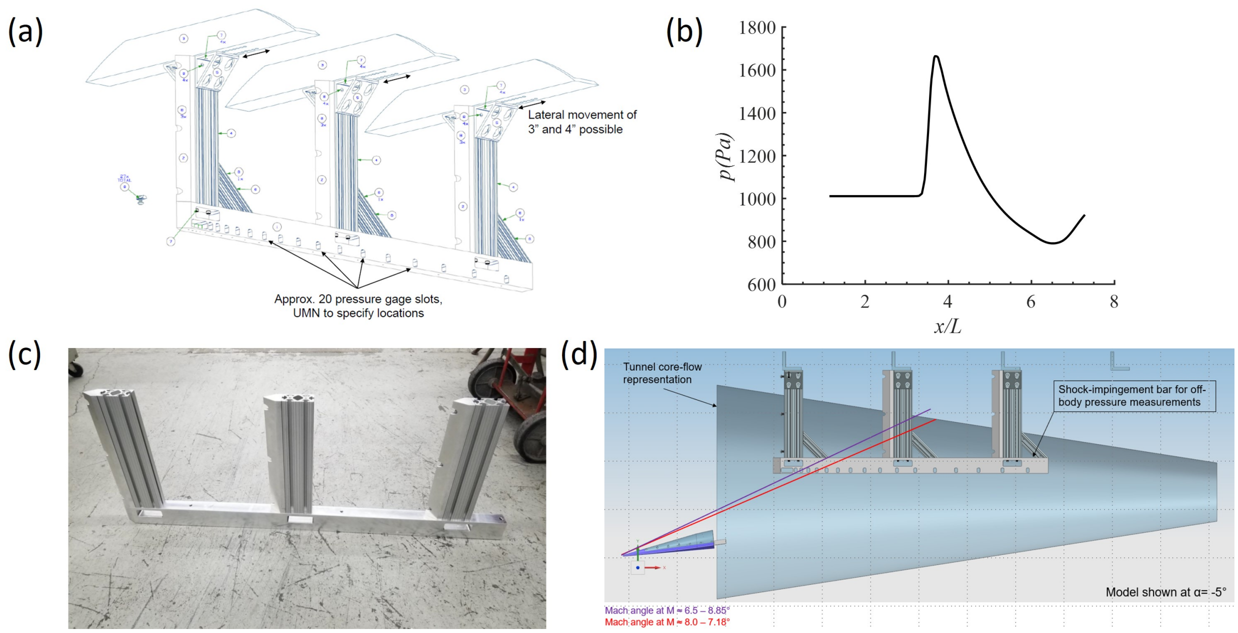

Fig. 6 (a) 3D view of the designed off-body pressure rail together with the mount in the tunnel to enable its lateral movements in the z direction. (b) The target signature obtained from CFD for the $M_\infty = 6.5$ and $\alpha = 0$ case. One could notice it is much smoother (but much larger amplitude) compared to signatures at low supersonic speeds. (c) Fabricated pressure rail in CUBRC machine shop. (d) The fourth mounts to enable the flexible movements in the streamwise direction. With marks for tunnel core section and shock angle, it is seen that the interference with reflected wave and boundary layer are avoided.

For different pitch angle α, three different sting blocks are then fabricated with α values of 0°, +5°, and −5°. The sting block structure and the final installation of the test article are given in Fig. 7.

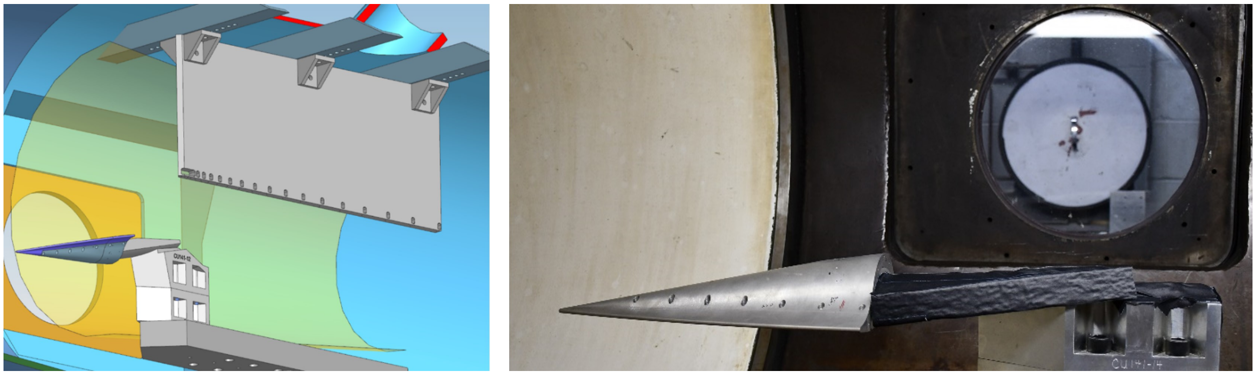

Fig. 7 The conceptual illustrations of the installed in-tunnel configuration (left); and the photo of the installed test article in the shock tunnel (right).

B. CFD simulation tools

The well-verified US3D solver [20–22] was used to conduct the CFD simulations. To account for the potential laminar-to-turbulent boundary layer transition with low computational cost, both laminar and RANS (using the Spalart-Allmaras model) simulations were performed to predict the boundary layer before and after the transition, respectively. The isothermal wall condition with $T_w = 300$ K was used, justified by the relatively short running time. The exact simulation conditions are the same as the shock tunnel test conditions, which are presented in Table 2. Second-order central scheme spatial discretization and first-order implicit time integration were employed. A mesh with about 8 million cells for half of the geometry are used. It takes about 170,000 iterations to reach the steady state. Limited by the design requirement, the θ and r coordinate is translated to off-track distance Δy and offset distance from the symmetry plane Δz (see Fig. 1). Pressure signatures were sampled at various locations.

The grid topology for CFD simulation was generated by the GoHypersonic LINK-3D grid generation software. With the consideration of potential impact from the sting holder on the wake, the sting holder is also included in the computational domain. The geometry with the sting holder is shown in Fig. 2. A half of the geometry is utilized to account for the planned pitch angle cases from $-5°$ to $5°$. The outer boundaries of the domain is specifically enlarged to allow more sampling space for the pressure signatures at large radii.

IV. Results and discussion

The objective of this study is to validate the CFD simulations by experimental measurements for a series of shock tunnel runs (Sec. IV.A). For this reason, various targets are picked for a systematic validation. There targets includes Force balance (Sec. IV.B), surface heat transfer and pressure (Sec. IV.C), shock angle (Sec. IV.D), and off-body pressure signature (Sec. IV.E). Estimation of the measurement uncertainty was made in each section.

A. List of shock tunnel test runs

The list of shock tunnel runs are shown in Table 2. Two flight Mach numbers $M_\infty = 6.5$ (the lower limit of the CUBRC's LENS 48" Shock Tunnel) and $M_\infty = 8$ (the upper limit of DARPA's interested Mach number range for this project) are selected. For each free-stream Mach number, three different pitch angles are tested ($\alpha = +5°$, $\alpha = 0$ and $\alpha = -5°$). For each combination of free-stream Mach number and pitch angle, two different location ($z = 0$ and $z = 4$ inch) are sampled from two different runs with fixed nominal free-stream condition. The target test condition for all $M_\infty = 6.5$ cases is $\rho_\infty = 1.29 \times 10^{-4}$ (slug/ft^2), $p_\infty = 0.2$ psia, and $Re_\infty = 4.71 \times 10^6$ ft^{-1}. The target test condition for all $M_\infty = 8$ cases is $\rho_\infty = 3.43 \times 10^{-5}$ (slug/ft^2), $p_\infty = 0.05$ psia, and $Re_\infty = 1.54 \times 10^6$ ft^{-1}. By comparing the as run conditions with the target, one could notice that the approximate 30% off of the free-stream unit Reynolds number compared with the target value. This variation is caused by the lower free-stream density in all test cases. The lower free-stream Reynolds number might suppress the boundary layer transition and finally results in some fully laminar states observed in the on-body measurements (will be discussed in Sec. IV.C).

Table 2 List of conditions for the shock tunnel test runs. Specifically, Re_∞ (ft^{-1}) is the free-stream unit Reynolds number (i.e., Re defined by by unit length without choosing a characteristic length scale).

Run #	As run M_∞	Lateral offset Δz (inch)	Pitch angle α	p_∞ (psia)	ρ_∞ (slug/ft^2)	Re_∞ (ft^{-1})
4	8.0	0	0	0.04	3.04×10^{-5}	1.12×10^6
5	6.6	0	0	0.15	9.45×10^{-5}	2.79×10^6
6	6.6	0	5	0.14	9.10×10^{-5}	2.67×10^6
7	8.0	0	5	0.04	3.09×10^{-5}	2.67×10^6
8	8.0	4	5	0.04	2.76×10^{-5}	1.02×10^6
9	6.6	4	5	0.15	9.93×10^{-5}	2.94×10^6
10	6.6	4	0	0.15	9.58×10^{-5}	2.84×10^6
11	8.0	4	0	0.05	3.11×10^{-5}	1.15×10^6
12	8.0	0	-5	0.05	3.15×10^{-5}	1.17×10^6
13	6.6	0	-5	0.14	9.32×10^{-5}	2.77×10^6
14	6.6	4	-5	0.15	9.21×10^{-5}	2.67×10^6
15	8.03	4	-5	0.04	3.06×10^{-5}	1.13×10^6

B. Force balance

The force values obtained from the balance measurements and CFD simulations are then compared, as shown in Table 3. Six cases, based on free-stream Mach number M_∞ and pitch angle α, were chosen from this comparison. Where the subscript "L" of the forces indicates the results obtained from the simulation with laminar flow assumption, and subscripts "R" of the forces indicates the results obtained from RANS simulations. Based on the comparison, at these test conditions, the potential transition to turbulent boundary layer does not contribute significantly in global forces. We only noticed very small prediction error in most of the force prediction from CFD other than the lift of Run #9, for which the lift is close to zero and hence the relative uncertainty is expected to be higher.

Table 3 Force validation by cases: D_m is the measured drag, and L_m is the measured lift; subscript "L" of the forces indicates the results obtained from the CFD simulation with laminar flow assumption, and subscripts "R" of the forces indicates the results obtained from the RANS simulations.

Run #	D_m (N)	D_L (N)	D_R (N)	L_m (N)	L_L (N)	L_R (N)
7	12.054	10.815	13.042	5.404	2.212	2.346
9	27.533	23.582	30.399	0.774	7.219	6.886
10	35.117	36.980	41.912	99.053	102.914	105.997
11	18.886	16.337	18.394	58.989	42.943	43.969
12	25.496	22.972	26.662	120.879	90.717	92.924
13	46.864	53.309	59.765	212.561	216.995	219.967

C. Surface heat transfer and pressure: laminar-to-turbulent boundary layer transition

In this section, the surface heat transfer and pressure sensor data are sampled to validate the CFD simulations. Since not all sensors are aligned, only the center line data are compared. Depending on whether the laminar or RANS simulation agrees with the experimental measurements, we can determine whether the laminar-to-turbulent boundary layer transition occurs and estimate the possible transition location. To enhance the confidence of the estimation, surface distribution of the dimensionless number Re_θ/M_e (a common empirical transition indicator of the start of linear growth) is shown in Fig. 8. Key findings from all these results are discussed below for each shock tunnel run.

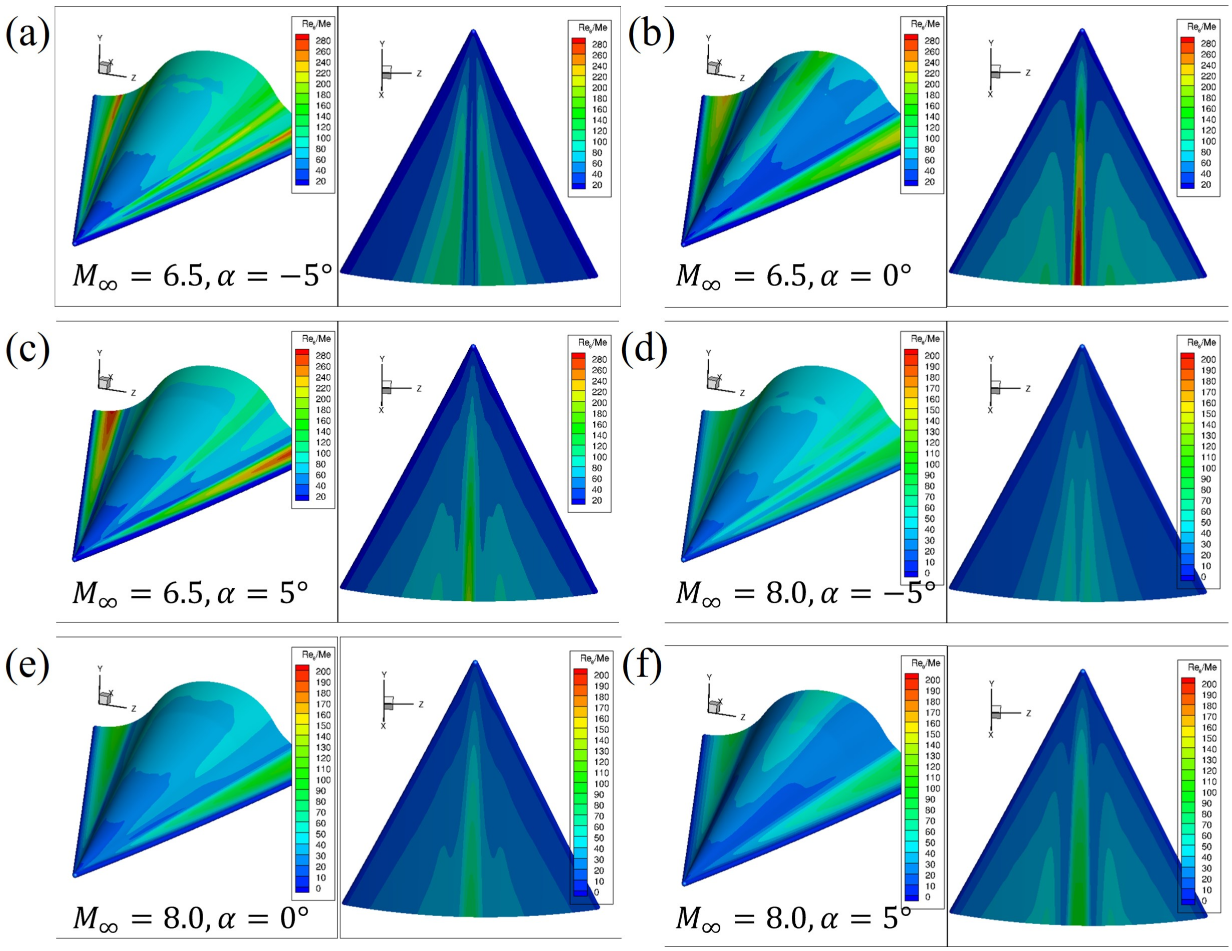

Fig. 8 Surface distribution of Re_θ/M_e calculated from CFD simulations to predict laminar-to-turbulent boundary layer transition and estimate a boundary layer's susceptibility to tripping the flow. The corresponding free-stream conditions are: (a)$M_\infty = 6.5$, $\alpha = -5°$; (b)$M_\infty = 6.5$, $\alpha = 0°$; (c)$M_\infty = 6.5$, $\alpha = 5°$; (d)$M_\infty = 8.0$, $\alpha = -5°$; (e)$M_\infty = 8.0$, $\alpha = 0°$, and (f)$M_\infty = 8.0$, $\alpha = 5°$.

1. Run 7 ($M_\infty = 8.0$, $\alpha = 5°$)

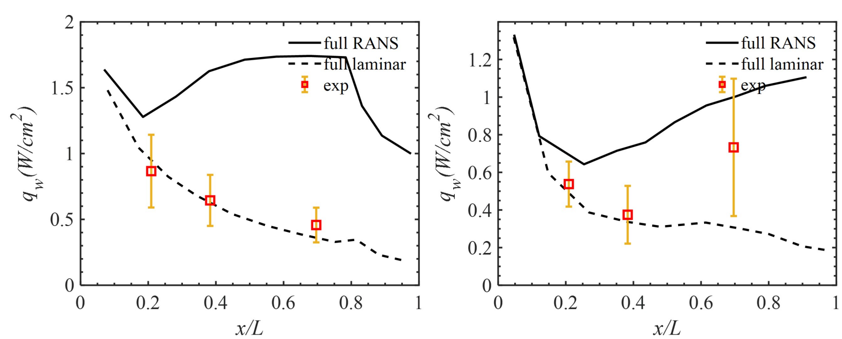

Fig. 9 The measured heat transfer flux q_w data compared to CFD simulation data for test run 7 ($M_\infty = 8.0$, $\alpha = 5°$). Left: top surface center line; Right: bottom surface center line.

The on-surface sampling results for run 7 ($M_\infty = 8.0$, $\alpha = 5°$) are plotted in Fig. 9. The measured heat transfer flux q_w are compared with the simulated results at the center line. The pressure comparison is not available since both pressure sensors on the top surface were damaged during the shock tunnel test. On the top surface, all of the three sensors perfectly match with laminar simulation, which indicates a very delayed or no transition. On the bottom surface, there might be a start of instability growth happened between $x/L = 0.4$ and 0.7 or after $x/L = 0.7$. Taking the large uncertainty encountered by the third sensor into account, no transition is also possible. This observation could be confirmed by Fig. 8(f), where Re_θ/M_e of the center line on both top and bottom surface are lower than 200.

2. Run 9 ($M_\infty = 6.5$, $\alpha = 5°$)

The on-surface center line heat transfer flux q_w and pressure p_w data for run 9 ($M_\infty = 6.5$, $\alpha = 5°$) are plotted in Fig. 10. The second pressure gauge on the bottom surface center line was found damaged in most of the tests. While the top surface is mainly maintained in laminar regime, the bottom surface boundary layer transits to turbulent between $x/L = 0.4$ and $x/L = 0.7$. The possible reason might be higher free-stream Mach number. Higher Re_θ/M_e close to 200 is also could be found at the center line on the bottom surface in Fig. 8(c). The pressure sensor data is off compared to both CFD predictions.

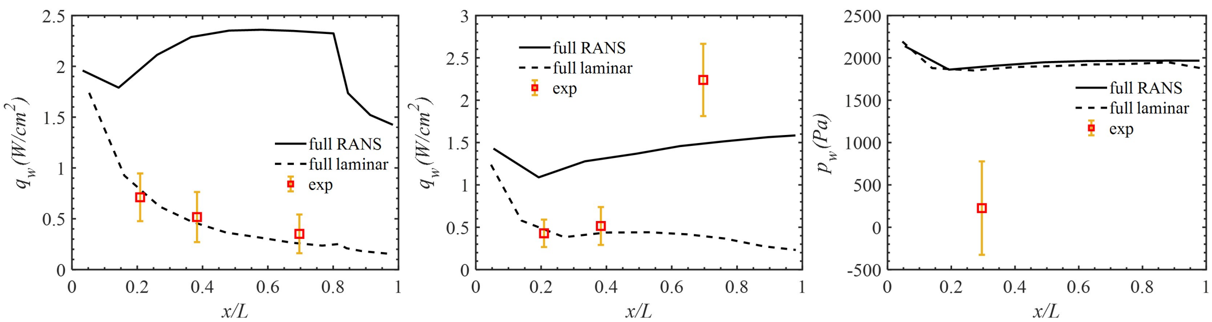

Fig. 10 The measured heat transfer flux q_w data and pressure p_w compared to CFD simulation results for test run 9 ($M_\infty = 6.5$, $\alpha = 5°$). Left: top surface center line q_w; Middle: bottom surface center line q_w; Right: bottom surface center line p_w.

3. Run 10 ($M_\infty = 6.5$, $\alpha = 0°$)

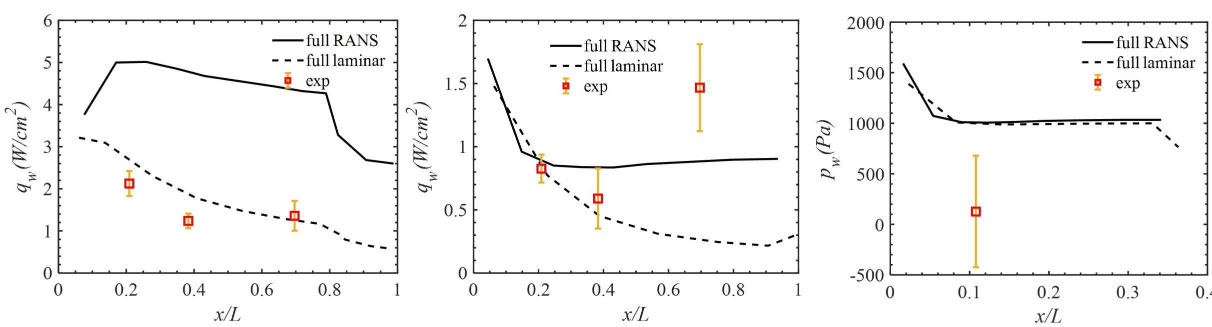

Fig. 11 The measured heat transfer flux q_w and pressure p_w data compared to CFD simulation results for test run 10 ($M_\infty = 6.5$, $\alpha = 0°$). Left: top surface center line q_w; Middle: bottom surface center line q_w; Right: bottom surface center line p_w.

The on-surface center line results for run 10 ($M_\infty = 6.5$, $\alpha = 0°$) are plotted in Fig. 11. While the top surface still remains laminar, the bottom surface boundary layer transits to turbulent between $x/L = 0.4$ and $x/L = 0.7$. This could be confirmed by the CFD results in Fig. 8(b), where Re_θ/M_e is fount to be greater than 280 on the later half of the center line on the bottom surface, but only $\sim$ 180 at the top surface. Note that the as run Reynolds number is about 30%

lower than the target, and hence it is reasonable not to observe the transition on the top surface. The pressure sensor data on the top surface is slightly off compared to both CFD predictions.

4. Run 11 ($M_\infty = 8.0$, $\alpha = 0°$)

The on-surface center line results for run 11 ($M_\infty = 8.0$, $\alpha = 0°$) are plotted in Fig. 12. Interestingly, based on the heat transfer q_w data, the center line of both top and bottom surfaces transit to turbulent very early, possibly near the nose. In addition, the pressure sensor data matches with RANS simulation pretty well, which also confirms the transition to turbulence. Since Figure 8(e) indicates very low Re_θ/M_e value across all the surface, non-linear growth (which could result in critical Re_θ/M_e drop to about 100) might be responsible for the transition in this run.

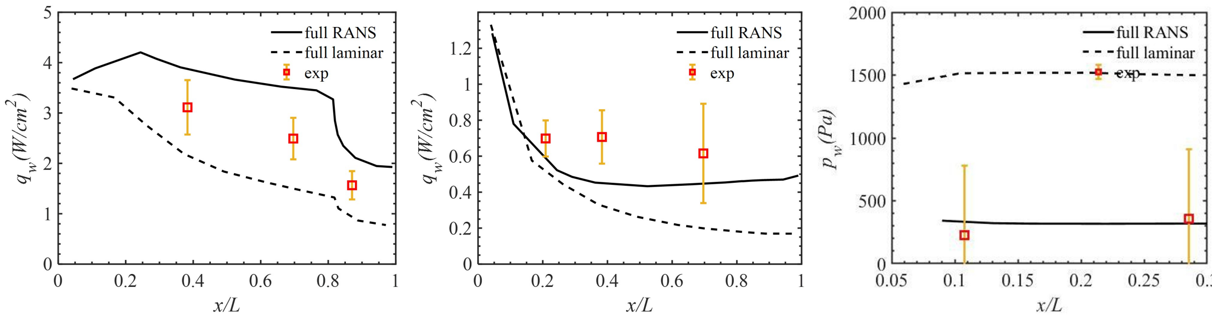

Fig. 12 The measured heat transfer flux q_w and pressure p_w data compared to CFD simulation results for test run 11 ($M_\infty = 8.0$, $\alpha = 0°$). Left: top surface center line q_w; Middle: bottom surface center line q_w; Right: bottom surface center line p_w.

5. Run 12 ($M_\infty = 8.0$, $\alpha = -5°$)

The on-surface center line results for run 12 ($M_\infty = 8.0$, $\alpha = -5°$) are plotted in Fig. 13. Unlike the findings in run 11, it seems that both top and bottom surface maintain laminar in this run. The predicted surface pressure agree with the measured data well, but the measurement uncertainty is bit large, which is caused by relative large scale range of the pressure gauges (15 psia).

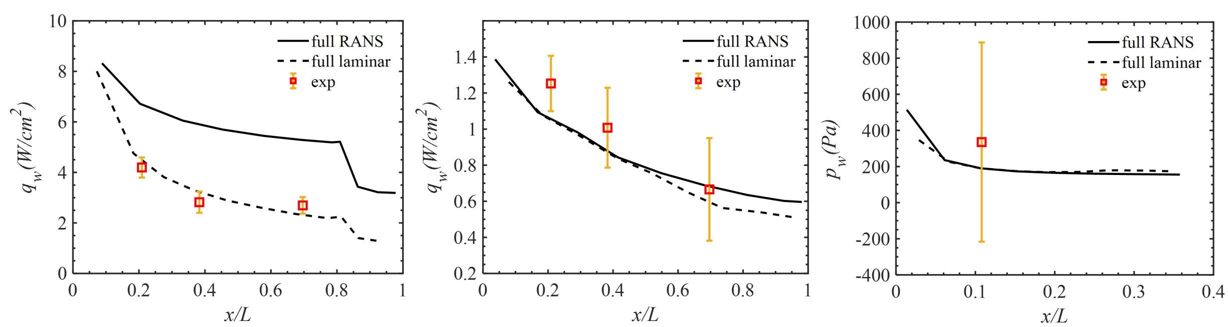

Fig. 13 The measured heat transfer flux q_w and pressure p_w data compared to CFD simulation results for test run 12 ($M_\infty = 8.0$, $\alpha = -5°$). Left: top surface center line q_w; Middle: bottom surface center line q_w; Right: bottom surface center line p_w.

6. Run 13 ($M_\infty = 6.5$, $\alpha = -5°$)

The on-surface center line results for run 13 ($M_\infty = 6.5$, $\alpha = -5°$) are plotted in Fig. 14. The center line of top surface starts transiting to turbulent between $x/L = 0.4$ and $x/L = 0.7$. The bottom surface center line mainly stays in laminar. The top surface Re_θ/M_e seems also relatively low based on the CFD transition analysis in Fig. 8(a), and hence, possibly some non-linear growth or instabilities by the free-stream are engaged. The laminar status of the bottom surface also matches with the CFD transition analysis since $Re_\theta/M_e < 100$ along it. The on-surface pressure agree with both CFD predictions well, but with a large uncertainty.

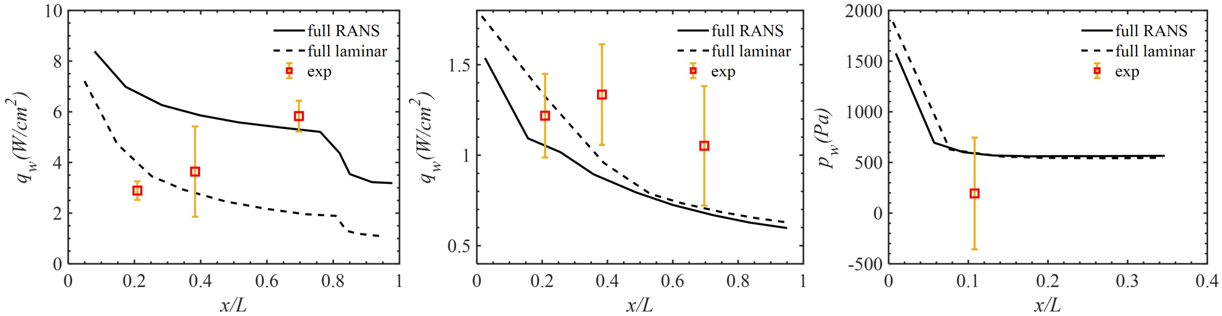

Fig. 14 The measured heat transfer flux q_w and pressure p_w data compared to CFD simulation results for test run 13 (M_∞ = 6.5, α = −5°). Left: top surface center line q_w; Middle: bottom surface center line q_w; Right: bottom surface center line p_w.

D. Schlieren image: shock angle

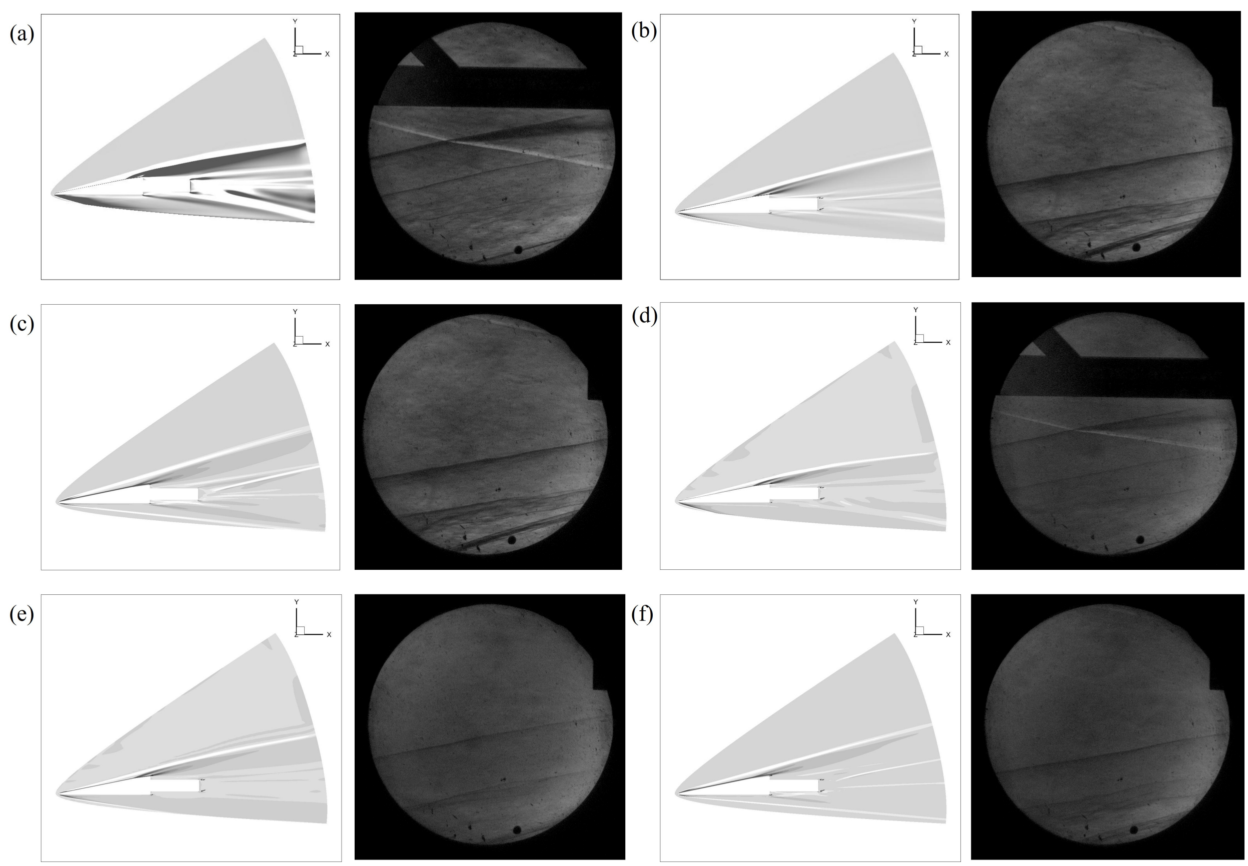

Fig. 15 Schlieren comparison for the shock tunnel runs. The corresponding free-stream condition are: (a)$M_\infty = 6.5$, $\alpha = -5°$; **(b)**$M_\infty = 6.5$, $\alpha = 0°$; **(c)**$M_\infty = 6.5$, $\alpha = 5°$; **(d)**$M_\infty = 8.0$, $\alpha = -5°$; **(e)**$M_\infty = 8.0$, $\alpha = 0°$; **and (f)**$M_\infty = 8.0$, $\alpha = 5°$**.**

The comparison of shock structures by schlieren images between CFD simulation (numerical schlieren: density gradient contours) and experiments are given in Fig. 15, which shows qualitative agreement on the incident shock. The shock-shock interaction caused by the interaction between the shock caused by the rail plate and the bow shock is observed, which might cause a deceleration of the gas flow and hence might result in pressure rise to pollute the off-body pressure measurement.

One could also perform quantitative comparison of the shock angle, as shown in Table 4. The measured shock angle also achieved quantitative agreement.

Table 4 Shock angle β comparison between experimental schlieren image and CFD prediction (based on numerical schlieren: i.e., density gradient).

Run #	M_∞	α (°)	β (°)	β_{CFD} (°)
7	8.0	5	15.0	14.5
9	6.5	5	14.5	15.5
10	6.5	0	9.0	11.5
11	8.0	0	10.5	10.5
12	8.0	-5	11.0	11.0
13	6.5	-5	12.5	12.5

E. Off-body pressure signature (sonic boom signature)

Prior to the off-body validation, the sensitivity of off-body pressure signature to the potential transition (laminar vs. RANS) is tested at $M_\infty = 6.5$ and $\alpha = 0$ (i.e., run 10), and the pressure signature was sampled at $r = 0.1$ m. Based on Fig. 16, the off-body signature only shows about 10% difference in the peak over-pressure Δp_{max} value and collapse to each other at other location except for the tail, which indicates the off-body signature is not very sensitive to the status of the flow (i.e., laminar or turbulent). For this reason, only laminar CFD results are compared to the experimental measurements.

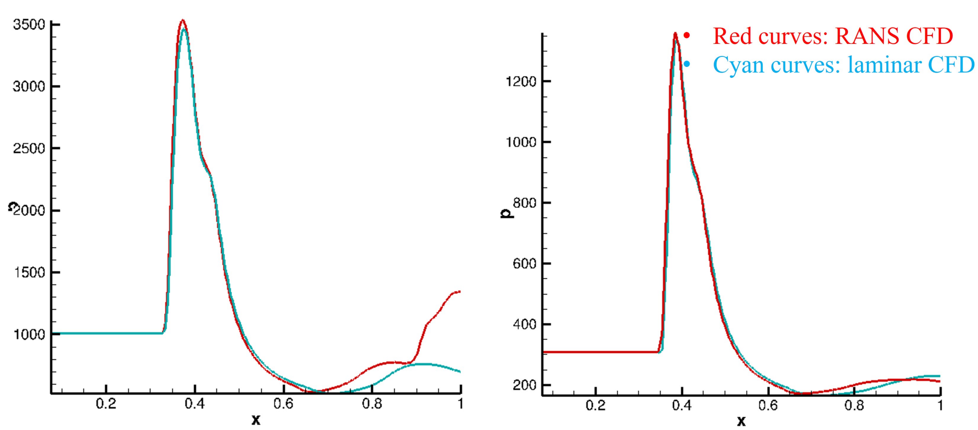

Fig. 16 Comparison between the laminar and RANS simulations in terms of the off-body pressure signature at $r = 0.1$ m for run 10 ($M_\infty = 6.5$ and $\alpha = 0$). Left: $z = 0$; Right: $z = 4$ inch.

We compared the off-body measurement of run 10 / run 5 ($M_\infty = 6.5$, $\alpha = 0°$ with $z = 4''$ / $z = 0$) and run 11 / run 4 ($M_\infty = 8.0$, $\alpha = 0°$ with $z = 4''$ / $z = 0$), which are shown in Fig. 17.

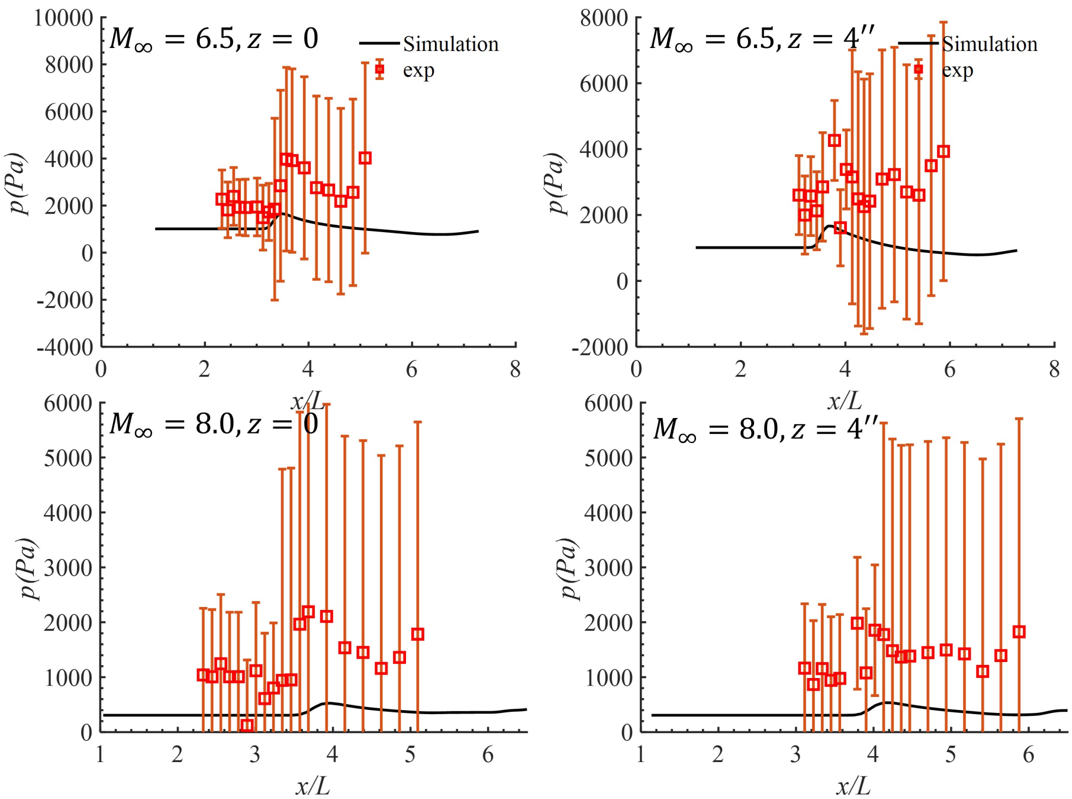

Fig. 17 Off-body pressure measurement compared with CFD. Pitch angle $alpha$ of both cases are 0. M_∞ and Δz are marked in the corresponding plots.

Although the mean value obtained by pressure signature are much higher than the CFD prediction, the CFD prediction is still within the measurement uncertainty range. The large measurement uncertainty found here is mainly contributed by the 0.8% measuring range system error accompanied by large measuring range of the pressure gauges used in this series of shock tunnel tests (15 or 50 psia: equivalently, about 103, 421 or 344, 738 Pa).

V. Conclusion

First tunnel tests for hypersonic sonic boom of a slender body was performed at different free-stream Mach numbers and pitch angles. The CFD simulations are validated by the measurement data through various perspectives, including force balance measurement, surface pressure and heat transfer (or, in another word, boundary layer transition) measurement, shock angle measurement (by schlieren images), and off-body pressure measurement. Through the comparison between the CFD prediction and measured surface pressure and heat transfer data, it could be inferred that the anticipated boundary layer transition takes place in some of the cases. By comparing the RANS and laminar simulation data at same free-stream condition, it is found that the impact of natural transition to turbulent boundary layer has very limited impact on the off-body pressure signature. Through schlieren images, it is found that the current pressure rail will trigger shock-shock interaction, which may distort the measured off-body pressure. For this reason, the off-body pressure rail might need further revision or new methodology to eliminate or minimize the impact of the presented shock-shock interaction. The current off-body measurement has large uncertainty, which is caused by the too large measuring range of the pressure gauges. For this reason, new pressure gauges with smaller measurement ranges are required for future shock tunnel tests to avoid burying measuring targets in noise.

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